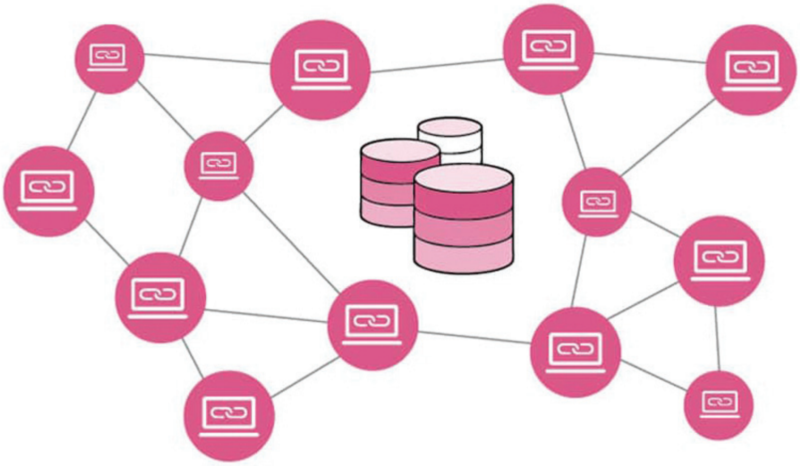
Blockchain Based Databases: Decentralised and Secure



In traditional databases, the centralised server maintains the complete set of tables and records. Such servers are subject to malicious attacks including SQL injections, cross-site scripting, and so on. Blockchain based databases are decentralised and are therefore very secure. This article introduces you to some of the best blockchain based databases.

Almost all government and corporate services are today available on digital platforms and smart gadgets. These online services and platforms are vulnerable to cyber-attacks and data leakage issues. This is because centralised database servers can be hacked by third parties and malicious traffic.

To meet this security challenge, blockchain based technologies are under development for multiple applications including e-governance, Internet of Things (IoT), online games, smart health services, supply chain management, and so on. Blockchains provide a decentralised digital database for storage of data with a higher degree of security and privacy. Cryptocurrencies like Bitcoin, Ethereum, Dogecoin, Stellar and many others rely on blockchains to keep track of transactions in a safe and decentralised manner. A blockchain ensures data accuracy and security, and can be trusted. No trusted third party is needed. Each node on a computer network has a copy of the blockchain's database.

Key applications and use cases of blockchain

- Cryptocurrencies
- Smart contracts
- Financial services
- Gaming and entertainment
- Supply chain management
- Medical and healthcare
- E-governance
- Media and advertising
- Telecommunications
- Manufacturing
- Travel and transportation
- Social media

Blockchain databases: What's different?

The blockchain is a decentralised database or public ledger that records all network transactions. Data in a database is typically organised into tables, but data in a blockchain is organised into blocks (as per the name) that are linked together without scope for manipulation or hacking. Using this data structure in a decentralised manner creates an irreversible chronology of data which provides a very high level of security, integrity and privacy. Once a block has been linked in the blockchain, it becomes permanent and nobody can alter it accidentally or intentionally.

A blockchain organises data into blocks, and each of these contains a specific sequence with encrypted data. The blockchain is a chain of data made up of blocks, and these store a set amount of data before being closed